INCH-POUND
MIL-PRF-83383/2G
w/AMENDMENT 3
19 December 2019
SUPERSEDING
w/AMENDMENT 2
MIL-PRF-83383/2G
4 February 2019

PERFORMANCE SPECIFICATION SHEET

CIRCUIT BREAKERS, REMOTE CONTROL, AC-DC, THERMAL, TRIP-FREE, SERIES TRIP, SINGLE POLE, AUXILIARY CONTACTS (5 TO 100 AMPERES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83383.

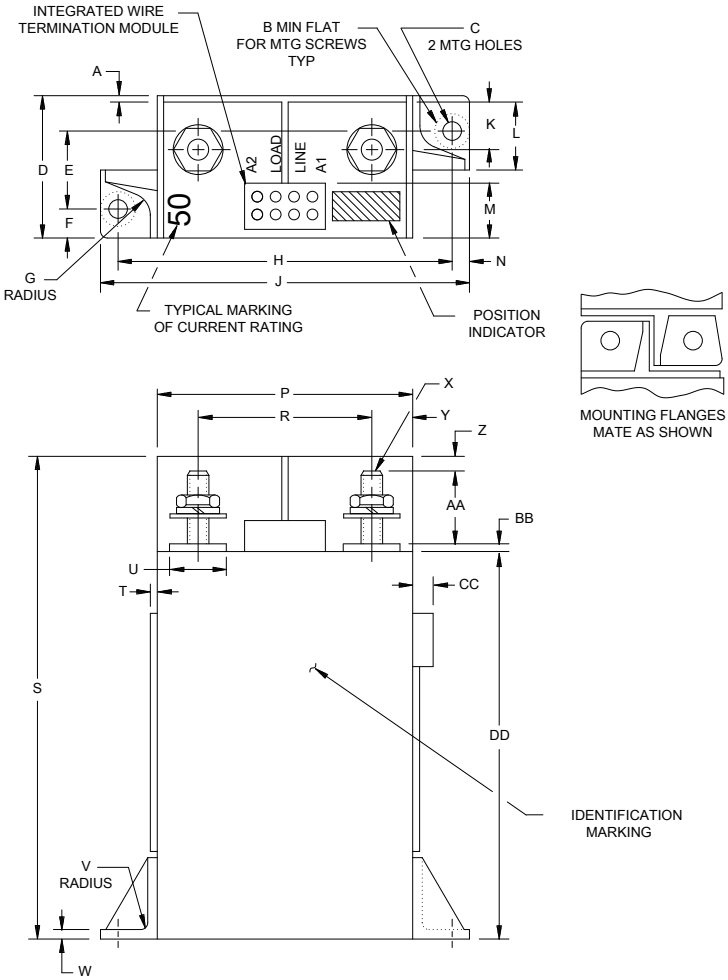


FIGURE 1. Interface and physical dimensions.

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Dimensions.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.056	---	1.42
B	.400	---	10.16	---
C	.167	.177	4.24	4.50
D	---	1.256	---	31.90
E	.673	.703	17.09	17.86
F	.241	.271	6.12	6.88
G	.200	---	5.08	---
H	2.925	2.955	74.30	75.06
J	---	3.250	---	82.55
K	.390	.450	9.91	11.43
L	---	.600	---	15.24
M	.469	.499	11.91	12.67
N	.120	.180	3.05	4.57
P	---	2.250	---	57.15
R	1.515	1.545	38.48	39.24
S	---	4.260	---	108.20
T	---	.061	---	1.55
V	.030	.090	.76	2.29
W	.074	.094	1.88	2.39
Y	.320	.380	8.13	9.65
Z	.070	---	1.78	---
BB	.035	.065	.89	1.65
CC	---	.180	---	4.57
DD	3.390	3.450	86.11	87.63

Terminal-Dimensions and Hardware.

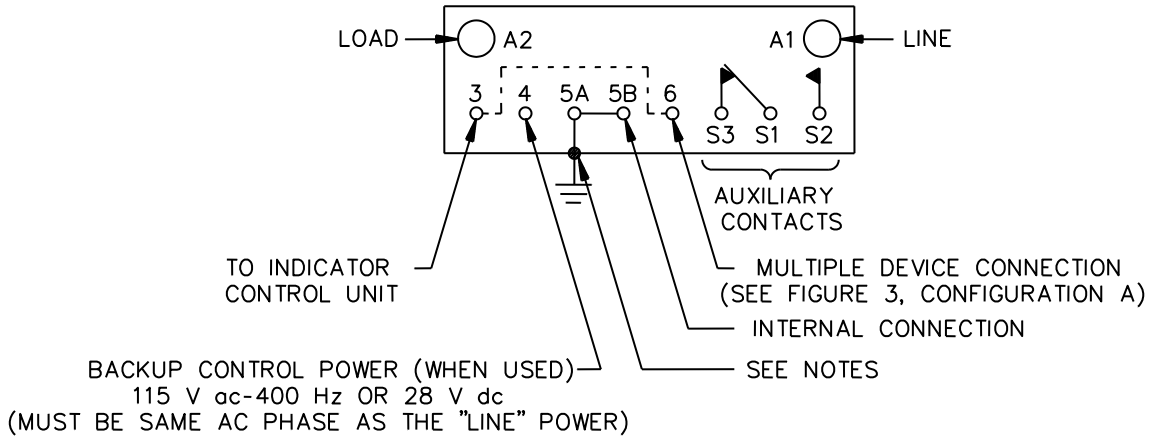
Current rating amperes	X	AA +.015 -.000	U	Nut 1/	Lock washer 2/	Flat washer 3/
5 to 25	.190-32UNC-2A	.500	.500	AN315-3R	MS35338-43	NAS1149F0332P
35 to 100	.250-28UNF-2A	.610	.610	AN315-4R	MS35338-44	NAS1149F0463P

1/ PIN is from National Aerospace Standard NASM315.
2/ PIN is from National Aerospace Standard NASM35338
3/ PIN is from National Aerospace Standard NAS1149.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm) for three decimal places and ±.03 (0.76 mm) for two decimal places.
4. Stiffening webs at mounting flanges shown for reference profile as required.
5. Current rating markings may be rotated 90 degrees from the type marking shown in order to accommodate large ampere ratings.

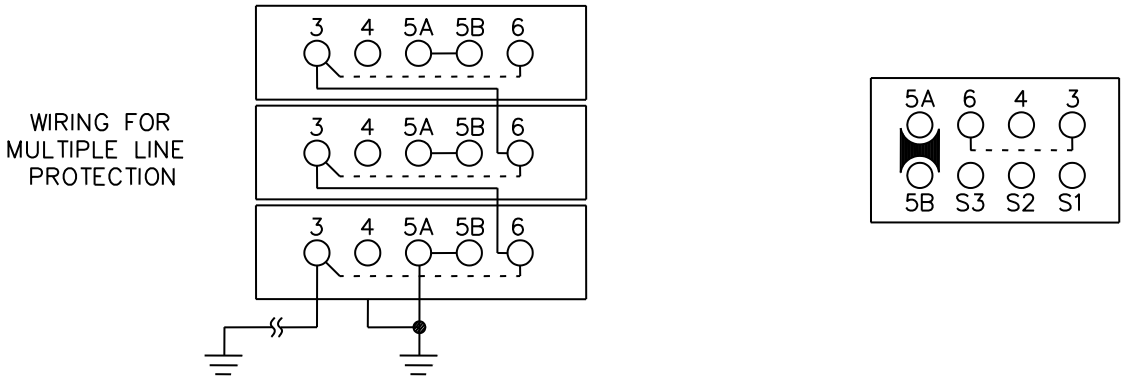
FIGURE 1. Interface and physical dimensions – Continued.

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NOTE: Terminals 5A and 5B shall be internally grounded to the mounting flange(s). Terminals 3 and 6 shall be internally connected.

FIGURE 2. Schematic.



NOTE: Terminal junction system (TJS) module shall accept pin contacts P/N M39029/1-100 or -101 as specified in SAE-AS39029/1.

FIGURE 3. Terminal junction system module.

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TABLE I. Dash numbers and characteristics.

Dash number	Current rating (amperes)	Tripping times from -54°C ±5°C to +71°C ±5°C (time in seconds)					
		Percent rated current					
		200		400		1000	
		Min	Max	Min	Max	Min	Max
01	5.0	7	40	1.2	6.4	.3	1.2
02	7.5	11	40	2.4	6.8	.33	1.1
03	10.0	12	42	2.8	8.5	.42	1.05
04	15.0	13	45	1.7	8.3	.35	1.2
05	20.0	14	46	2.9	7.6	.4	1.15
06	25.0	15	50	2.6	8.7	.4	1.3
07	35.0	16	55	2.8	8.3	.35	1.3
08	40.0	16	55	2.9	9.2	.36	1.3
09	50.0	13	55	2.9	10	.4	1.25
10	60.0	13	60	2.6	13	.26	1.8
11	75.0	13	60	2.5	13	.26	1.8
12	80.0	14	60	2.7	12.5	.3	2
13	100.0	17	63	3.5	13	.35	1.9

REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Weight:

5 to 25 amperes: 0.735 pound (334 grams), maximum.

35 to 50 amperes: 0.750 pound (341 grams), maximum.

60 to 100 amperes: 0.766 pound (348 grams), maximum.

Trip indication: See figure 1.

Terminals: See figures 1 and 2.

Auxiliary contact terminals:

Contact capacity at 28 V dc and 115 V ac, 400 Hz shall be 3 amperes resistive, 1.5 amperes inductive, and 0.5 ampere lamp load.

Voltage rating: 28 V dc or 115 V ac, 400 Hz.

Current ratings: See table I.

Tripping times: See table I.

Response time: 25 milliseconds, maximum.

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Power requirements and response time at minimum voltage:

Standby current drain and actuation current:

Test voltage: 28 ±0.5 V dc to TJS module, then 115 ±2 V ac, 400 Hz to line terminal. Actuating current, 3.5 amperes dc and 11 amperes ac rms maximum.

Maximum operating time at minimum voltage:

Test voltage: 18 V dc then 104 V ac rms, 400 Hz, maximum.

Endurance:

Electrical:

5 to 25 amperes: 50,000 cycles resistive or inductive or motor load or lamp load.

35 to 50 amperes: 50, 000 cycles resistive or inductive or motor load or 25,000 cycles lamp load.

60 to 100 amperes: 50,000 cycles resistive or inductive load or 25,000 cycles motor load.

Interrupting capacities: During the interrupt capacity test the circuit breaker may fail safe. For the purpose of this test, a circuit breaker, which has failed safe, is a part which is unable, because of electrical damage, to carry an electrical load. In addition, the leakage current between the load and the line terminals shall be less than 1 miliamp at 300 VAC with the circuit breaker in the closed or reset position. It is permissible to use rated cable for the interrupt capacity test once the calibrated bolted current is achieved. Length of the rated cable shall be 3 feet max.

A - 2,800 amperes.

B - 2,800 amperes.

C - 3,600 amperes.

D - 3,600 amperes.

Vibration-sinusoidal: Scan shall be conducted at 10 Hz to 55 Hz at .06 Double Amplitude (DA), 55 Hz to 2000 Hz at 10 G's with the frequencies varied logarithmically from 10 Hz to 2000 Hz (no return sweep), shall be traversed in 52 minutes ±5 minutes.

Post endurance vibration: Tests shall be run using 80 percent of rated load.

Other marking:

Current rating: See figure 1.

Line and load terminals: See figure 1.

Terminal junction system (TJS) module: See figure 3.

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Position indication: See figure 1. Indication is provided by a lens over “open” and “closed” on red and green background.

Part or Identifying Number (PIN): M83383/02- (dash number from table I) (e.g., M83383/02-01).

Indicator/control unit (I/CU): In order to maintain compatibility between the RCCB and the I/CU, the following must be observed when making an I/CU selection:

a. The I/CU must be a ½ ampere rated circuit breaker (MS22073-1/2 or MS26574-1/2, as specified in SAE-AS58091, or equivalent).

b. The line impedance (terminal 3 to I/CU) plus the I/CU impedance plus I/CU to ground connection shall not exceed 7.5 ohms.

c. The I/CU must respond to a decreasing current pulse of $I^2t \geq 2$ throughout the temperature range.

Referenced documents. In addition to MIL-PRF-83383, this document references the following:

NASM315 NASM35338 NAS1149 SAE-AS58091 SAE-AS39029/1

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Review activities:
Army - AT, CR4

Preparing activity:
DLA - CC

(Project 5925-2020-002)

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